

MATH METHODS FOR ECONOMISTS
COLUMBIA UNIVERSITY ECONOMICS DEPARTMENT
SYLLABUS FOR FALL 2026

Course Objective.

This course offers an introduction to mathematical tools and methods that are useful for the study of economics.

Course Structure.

We will meet Tuesdays and Thursdays, 11:40 am-12:55 pm in 608 Schermerhorn Hall. I will lecture twice weekly for the first half of the semester, up until October 15. Please see the detailed lecture schedule below. Following these lectures, we will have an in-class final exam.

Office Hours.

You may contact me at jenlao@columbia.edu

My Office Hours: Mondays 4:00-5:00 pm or by appointment in IAB 1014C.

Teaching Assistant.

The Teaching Assistant for this course is Bruno Escobar. He will hold office hours weekly and can answer questions about the material.

Bruno's email is be2329@columbia.edu

Bruno's Office Hours: TBA.

Textbooks.

There are no required textbooks for the course. Lecture notes will be based on material drawn from the following textbooks:

Rudin, Walter (1976). *Principles of Mathematical Analysis*. 3rd Edition. McGraw-Hill; New York, New York.

Stokey, Nancy L., Robert E. Lucas Jr., with Edward C. Prescott (1989). *Recursive Methods in Economic Dynamics*. Harvard University Press; Cambridge, Massachusetts.

Mas-Colell, Andreu, Michael D. Whinston, and Jerry R. Green, (1995). *Microeconomic Theory*. Oxford University Press.

Sundaram, Rangarajan K. (1996). *A First Course in Optimization Theory*. Cambridge University Press.

Ok, Efe A. (2007). *Real Analysis with Economic Applications*. Princeton University Press.

Hamilton, James D. (1994). *Time Series Analysis*. Princeton University Press.

Course Website.

All lecture notes, problem sets, and materials will be posted on the Courseworks website.

Problem Set Collaboration.

You are permitted and encouraged to collaborate with each other on the problem sets. However, in order to receive credit, each person must submit their own individual (separate) problem set.

Grading.

Your final grade in this course will be determined as follows:

80% Problem Sets

20% Final Exam

Lecture Schedule

1. Tuesday September 8
2. Thursday September 10
3. Tuesday September 15
4. Thursday September 17
5. Tuesday September 22
6. Thursday September 24
7. Tuesday September 29
8. Thursday October 1
9. Tuesday October 6
10. Thursday October 8
11. Tuesday October 13
12. Thursday October 15
13. Tuesday October 20: In-class Final Exam